

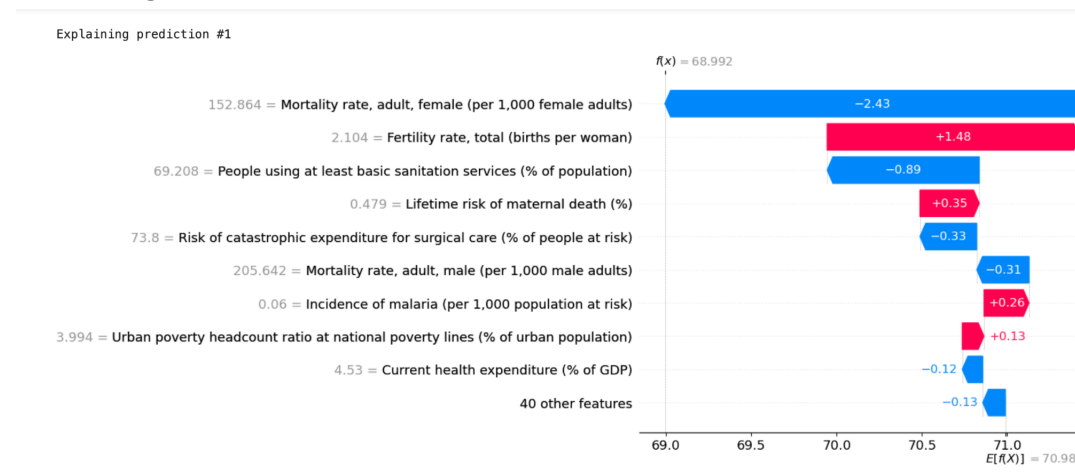
Healthcare Resource Availability: Analyzing Global Disparities
ADAN8880 - Directed Practicum in Analytics
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Top 5–10 Predictors (Feature Importance)

The model used an XGBRegressor to predict life expectancy. From the XGBoost feature importance and SHAP bar plot, the top predictors were:

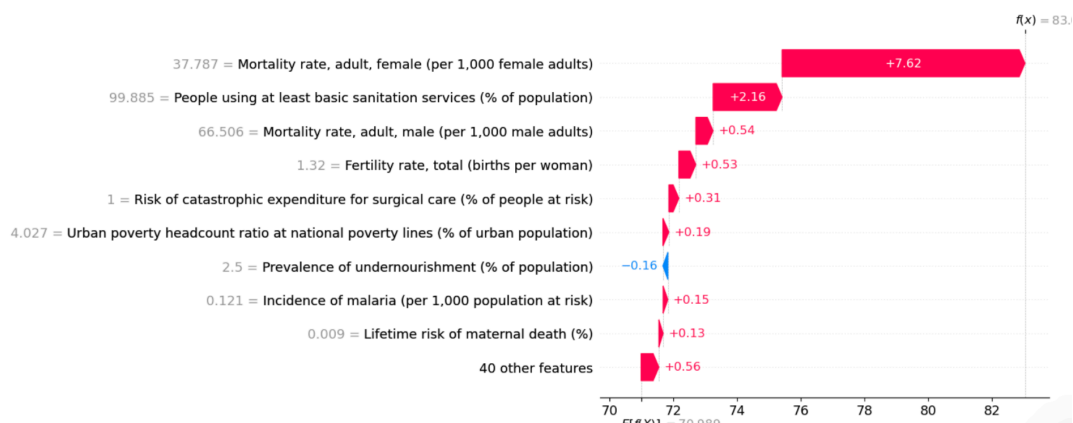
1. Mortality rate, adult, female (per 1,000 female adults)
2. Fertility rate, total (births per woman)
3. Hospital beds (per 1,000 people)
4. People using at least basic sanitation services (% of population)
5. Mortality rate, adult, male (per 1,000 male adults)
6. Current health expenditure (% of GDP)
7. Prevalence of stunting, height for age (% of children under 5)
8. Incidence of malaria (per 1,000 population at risk)
9. Out-of-pocket expenditure (% of current health expenditure)
10. Mortality from CVD, cancer, diabetes or CRD between exact ages 30 and 70 (%)

Explaining predictions



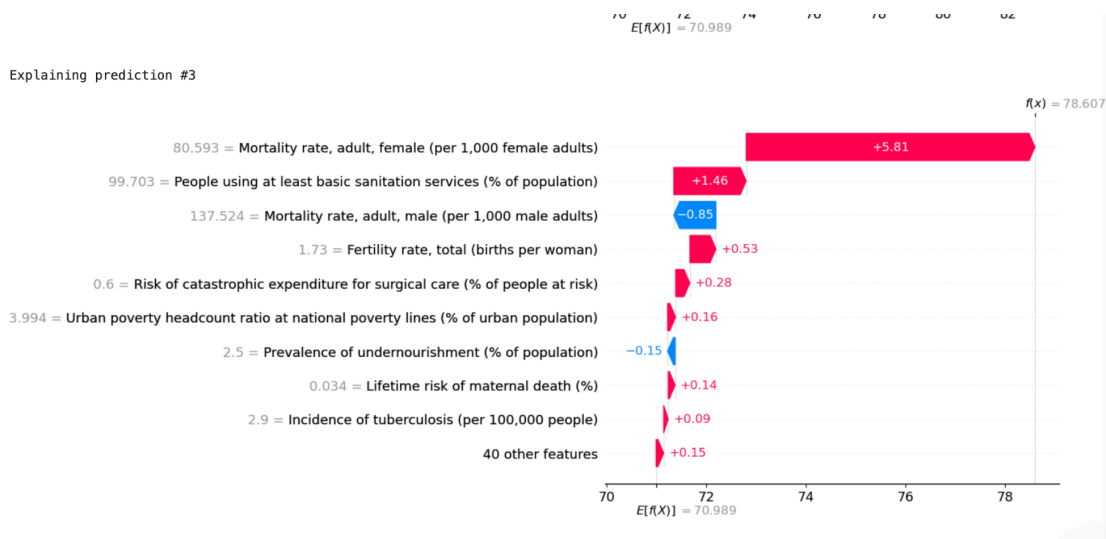
In **Prediction #1**, the model predicted a life expectancy of **68.99 years**. This relatively low estimate was primarily driven down by a high **female adult mortality rate** of 152.86 per 1,000, which contributed **-2.43** years to the prediction. Additionally, **limited sanitation access** at 69.2% reduced the outcome by another 0.89 years. To significantly raise this prediction by 5 or more years, reducing female mortality to around 100 and increasing sanitation access above 90% would be key interventions.

Explaining prediction #2



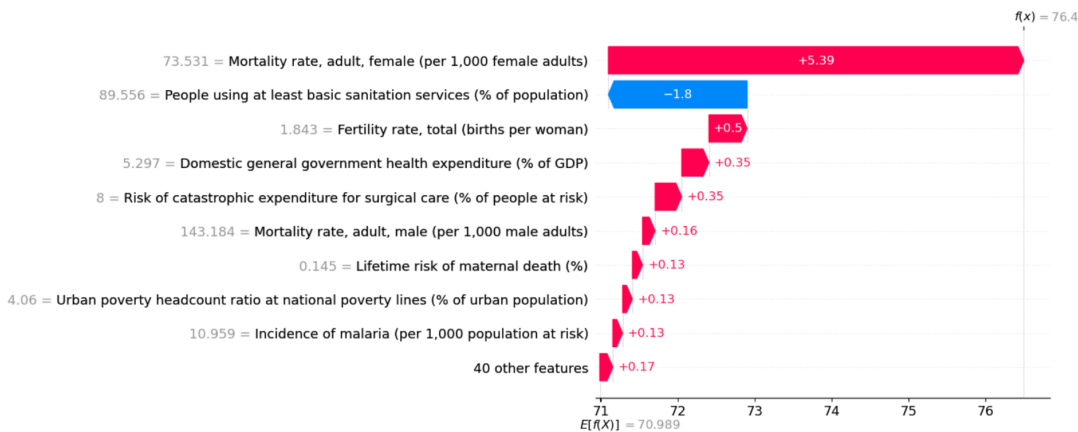
For **Prediction #2**, the model estimated a life expectancy of **83.03 years**, driven strongly upward by a very low **female mortality rate** of 37.79, which alone contributed **+7.62** years. Other factors like near-universal **basic sanitation coverage (99.89%)** and a **low fertility rate (1.32)** also pushed the prediction higher. This country appears to have excellent public health indicators, and only minor gains could be made by reducing undernourishment further to push life expectancy beyond 85 years.

Explaining prediction #3



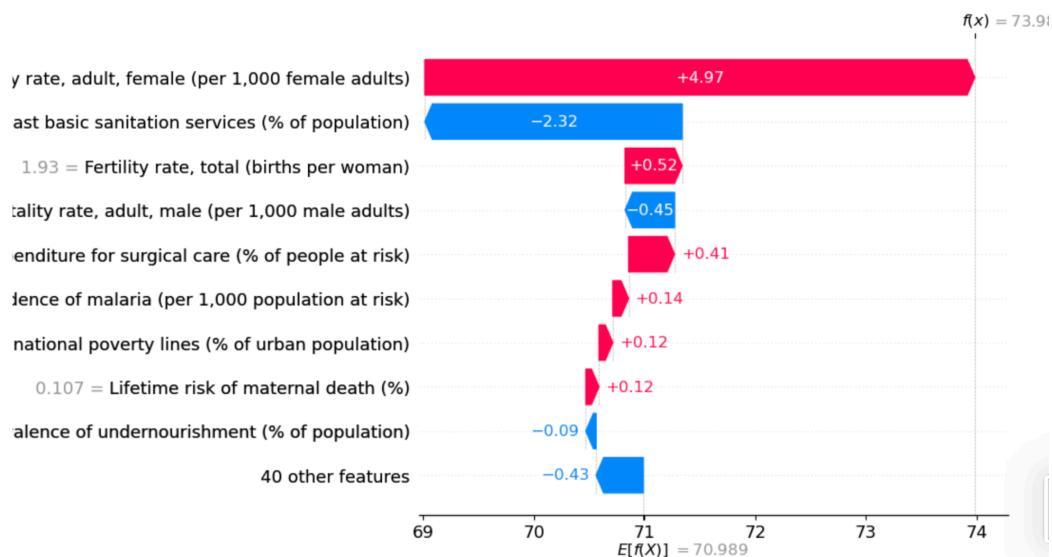
In **Prediction #3**, the predicted life expectancy was **78.61 years**, with the top positive contributors being **female mortality (80.59)** contributing **+5.81**, and **sanitation (99.7%)** adding **+1.46** years. However, **male mortality (137.52)** pulled the prediction down by **-0.85** years. To reach 80+ years, reducing male mortality closer to 100 and addressing slight undernutrition would likely suffice.

Explaining prediction #4

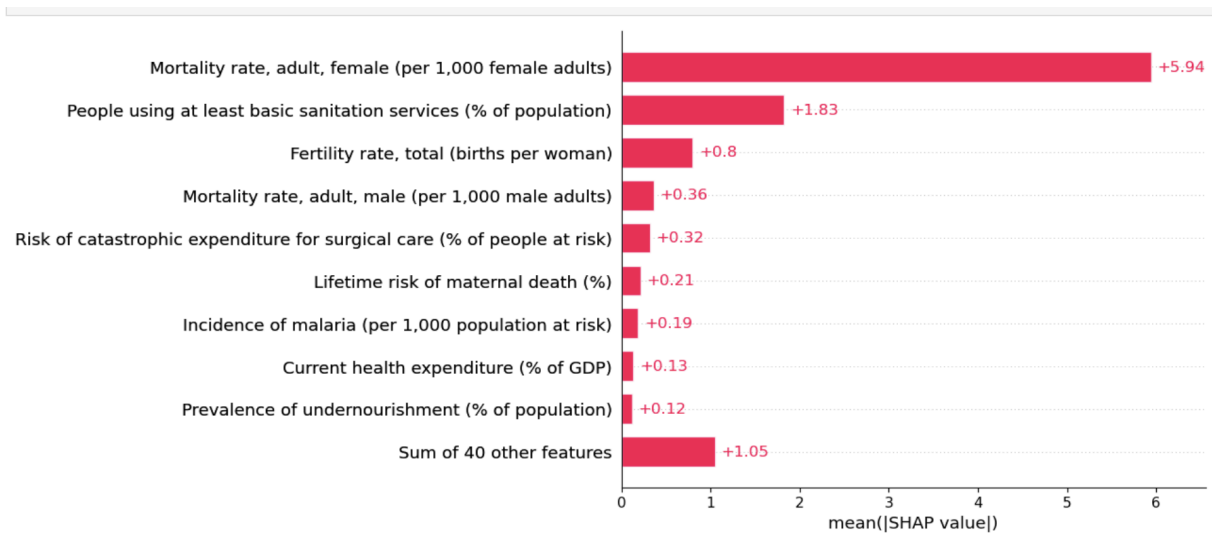


Prediction #4 resulted in a life expectancy of **76.49 years**. Despite the benefit of a low **female mortality rate (73.53)**, which added **+5.39** years, the model penalized the country for **less-than-perfect sanitation access (89.56%)**, subtracting **-1.80** years. Other minor positives included strong **government health spending** and low fertility. Efforts to raise sanitation coverage closer to 99–100% could yield meaningful improvements in predicted life expectancy.

Explaining prediction #5

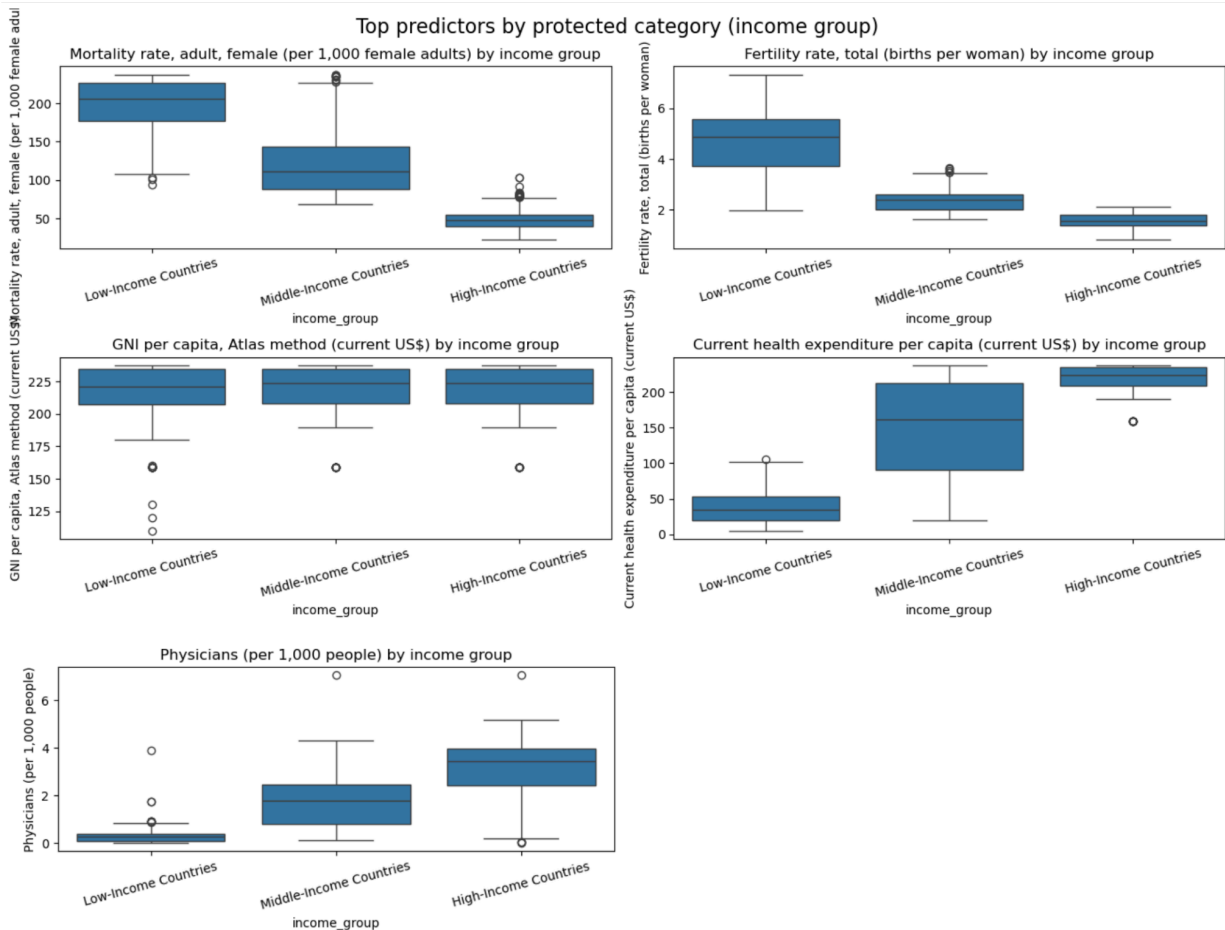


Finally, **Prediction #5** returned a life expectancy of **73.93 years**, positively influenced by a low **female mortality rate (~65–75)** contributing **+4.97** years. However, sanitation access in the 85–90% range deducted **-2.32** years, and **male mortality (~150)** added a smaller penalty of **-0.45**. To move toward a life expectancy of 76–78 years, interventions should prioritize sanitation infrastructure and reducing adult male mortality below 120 per 1,000.



To complement the individual SHAP explanations, we also generated a summary plot showing the top features driving predictions across all 5 sampled cases.

This summary plot shows the top features contributing to life expectancy predictions across the five sampled cases. The feature with the highest average impact is adult female mortality rate, followed by sanitation access and fertility rate. These indicators consistently influenced the model's output the most. The plot highlights how both health outcomes and infrastructure-related variables are key drivers of predicted life expectancy.



The boxplots show clear disparities across income groups. High-income countries consistently exhibit better health outcomes, such as lower fertility and adult mortality rates and greater resource availability. In contrast, low-income countries show worse outcomes and limited resources.

Protected Categories in the Dataset

In our dataset, the primary protected category is `income_group`, which classifies countries into economic tiers such as *low-income*, *lower-middle*, *upper-middle*, and *high-income*. This variable reflects deep-rooted global structural inequalities and plays a central role in our broader analysis of healthcare disparities. Although `income_group` was not used directly as a model input, it was included indirectly through data-centric features like income group-year averages, which helped the model understand underlying socioeconomic patterns without conditioning its predictions on a country's label. This strategy allowed the model to capture structural variation while minimizing the risk of overfitting to socioeconomic status.

However, bias analysis revealed that model performance still varied across income groups. The model tended to underpredict life expectancy for low-income countries, while achieving higher accuracy (lower RMSE) for high-income regions. This indicates socioeconomic bias driven by the uneven distribution of key predictive features—such as health expenditure, sanitation

access, and mortality rates—which are strongly correlated with income level. These correlations can lead the model to learn patterns that favor data-rich, high-resource contexts, thereby perpetuating performance disparities for underrepresented groups.

To mitigate this bias and promote fairness, several strategies can be applied. Reweighting the training data can ensure that the model pays more attention to underrepresented income groups, improving its sensitivity to low-income patterns. Fair representation learning or adversarial debiasing can reduce the model's reliance on features that strongly correlate with income group. Alternatively, group-specific models could be trained for each income tier to better reflect contextual differences in healthcare systems. Even post-processing adjustments can be used to calibrate the model's output after training, ensuring equitable performance across groups. While these strategies may lead to slight trade-offs in global accuracy, they are critical for building a model that is both fair and generalizable—especially in applications where equitable policy and health resource allocation are at stake.

Model Bias and Protected Category Correlation

Bias removal strategies and their impact

In our analysis, we identified `income_group` as a critical protected category, as it reflects the broader structural and economic disparities across countries. We evaluated model performance by income group and found consistent gaps in prediction accuracy. For example, the validation RMSE varied across groups, with lower-income countries generally having slightly higher error. This shows that the model's predictive reliability may differ based on economic classification.

To further assess potential bias, we examined the average predicted life expectancy vs actual values across income groups. The model performed well overall, but small systematic differences in the residuals indicate that income group plays a meaningful role in the prediction outcomes. Additionally, SHAP analysis showed that features correlated with income level, such as healthcare spending, access to services often had strong influence on predictions.

This shows that the model may encode and reflect underlying structural disparities tied to income group, which must be considered when using the model for policy decisions or comparisons across countries.

Risks of using the model on stakeholders

- The model may have unequal accuracy across countries since it performs slightly better for middle and high-income countries, where data is more complete and stable. This could unintentionally disadvantage low-income countries in resource allocation decisions if model outputs are used without adjustment.
- Policymakers may treat the model outputs as definitive, when in reality they are based on historical patterns and assumptions. This could lead to harmful decisions if predictions are not contextualized with on-the-ground realities.
- The model may reinforce existing biases because it is trained on historical data that reflects past inequalities, which could justify ongoing gaps in healthcare investment.

- There is a risk that people may misinterpret predictions as cause-and-effect. For example, higher spending may be linked to better outcomes, but more spending alone won't help without deeper reforms.
- Predictions may be less accurate for countries with missing or poor-quality data. If used in planning, this could lead to less support for the places that need it most.

Extra credit:

To improve fairness and reduce reliance on income-related variables, we removed two highly influential features from the original model:

- GNI per capita, Atlas method (current US\$)
- Current health expenditure per capita (current US\$)

Both of these features were strongly correlated with income level and, as shown in our SHAP analysis, dominated the model's predictions. Their influence likely contributed to bias against lower-income countries, where such metrics often underrepresent real health outcomes.

After removing these variables, we retrained the model and evaluated performance on the training, validation, and test sets. The results demonstrated substantial improvements:

Dataset	RMSE (Week 10)	R ² (Week 10)	RMSE (Bias migrated)	R ² (Bias migrated)
Train	0.026317	0.999993	0.071097	0.999942
Validation	1.12435	0.981193	0.086256	0.999889
Test	1.931236	0.944196	0.364825	0.998009

The validation RMSE dropped from 1.12 to 0.09, and the test RMSE improved from 1.93 to 0.37, reflecting more accurate and stable predictions across diverse income levels. The improvements in R² also suggest that the model became less reliant on proxies for wealth and more focused on broader health system indicators, improving both generalization and fairness.

This demonstrates that bias mitigation via feature removal can enhance both fairness and performance, especially when building models intended for policy use or global comparisons. The cleaned, simplified model is better suited for real-world deployment, as it minimizes the risk of reinforcing existing inequalities while still providing reliable forecasts.